

# HOLY ANGEL UNIVERSITY

## IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

A Capstone Project  
Presented to the Faculty of the  
School of Computing  
Holy Angel University



In Partial Fulfillment  
of the Requirements for the Degree  
Bachelor of Science in Computer Science

Quebec Student  
Romeo Student  
Ysabel Tolentino  
Zach Dizon

March 2026

*Proposal manuscript — draft (Introduction to Conceptual Framework). Later sections follow in the complete proposal.  
Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are  
placeholders.*

## HOLY ANGEL UNIVERSITY

### APPROVAL SHEET

This capstone entitled “**IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES**”, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science has been examined and is recommended for acceptance and oral examination.

\_\_\_\_\_  
**Dean Delta**  
Adviser

### ORAL EXAMINATION

Approved by the committee of oral examiners on  
[Date of defense]

\_\_\_\_\_  
**Prof. Foxtrot**  
Panel Chair

\_\_\_\_\_  
**Prof. Bravo**  
Panel Member

\_\_\_\_\_  
**[Name of Panel Member]**  
Panel Member

### APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science.

\_\_\_\_\_  
**Prof. Alpha**  
Program Chair/Coordinator,  
Computer Science

\_\_\_\_\_  
**Dean Delta**  
Dean, School of Computing

**HOLY ANGEL UNIVERSITY****Acknowledgment**

The researchers extend their sincere gratitude to everyone who supported this capstone project. This page is placeholder text: a real manuscript thanks the Adviser, the panel, the instructors, the host office and the families who made the work possible.

The group is especially grateful to Dean Delta for patient guidance through every revision.

# HOLY ANGEL UNIVERSITY

## Table of Contents

|                                                    | <b>Page</b> |
|----------------------------------------------------|-------------|
| Cover Page .....                                   | i           |
| Approval Sheet .....                               | ii          |
| Acknowledgment .....                               | iii         |
| Table of Contents .....                            | iv          |
| Title Page .....                                   | 1           |
| Introduction .....                                 | 2           |
| Review of Related Literature .....                 | 2           |
| Indoor Positioning with Bluetooth Low Energy ..... | 2           |
| Asset Tracking in Educational Laboratories .....   | 2           |
| Signal Smoothing and Classification .....          | 3           |
| Evaluating IoT Systems .....                       | 3           |
| Synthesis .....                                    | 3           |
| Conceptual Framework .....                         | 3           |
| Objectives of the Study .....                      | 3           |

**HOLY ANGEL UNIVERSITY****IoT-Based Laboratory Equipment Tracking for Computing Laboratories**

Quebec Student, Romeo Student, Ysabel Tolentino, Zach Dizon

Holy Angel University  
Computer Science

**Abstract**

*The abstract is written after the study is completed and is left blank at the proposal stage.*

**HOLY ANGEL UNIVERSITY****IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING  
LABORATORIES****Introduction**

The computing laboratories lend projectors, development boards, networking kits and spare peripherals to classes every day. Borrowing is recorded on paper, items are moved between rooms without a record, and the semestral inventory regularly finds equipment missing or in the wrong room.

Bluetooth Low Energy (BLE) tags are inexpensive and last months on a coin cell. Gateways in each room can detect the tags and report their signal strength, from which a server can estimate the room an item is in. Combined with a borrowing log, such a system would show where every item is and flag items that leave the laboratories.

The problem this study addresses is the following: laboratory staff cannot locate equipment without searching every room, borrowing records are incomplete, and losses are discovered only at inventory time. This capstone project therefore proposes the equipment tracker.

**Review of Related Literature**

This section reviews studies and existing systems related to the equipment tracker and the methods used to evaluate it.

***Indoor Positioning with Bluetooth Low Energy***

RSSI-based room-level positioning with one receiver per room achieves over 90% accuracy when readings are smoothed over a short window (Placeholder & Author, 2023).

***Asset Tracking in Educational Laboratories***

Tagged asset systems in school laboratories reduced search time and made borrowing records complete (Sample, 2024).

### ***Signal Smoothing and Classification***

Moving averages and Kalman filters reduce RSSI noise; a nearest-gateway rule with hysteresis prevents items from flickering between rooms (Example & Writer, 2022).

### ***Evaluating IoT Systems***

ISO/IEC 25010 is commonly applied to IoT prototypes alongside accuracy measurements of the sensing layer (Example, 2023).

### ***Synthesis***

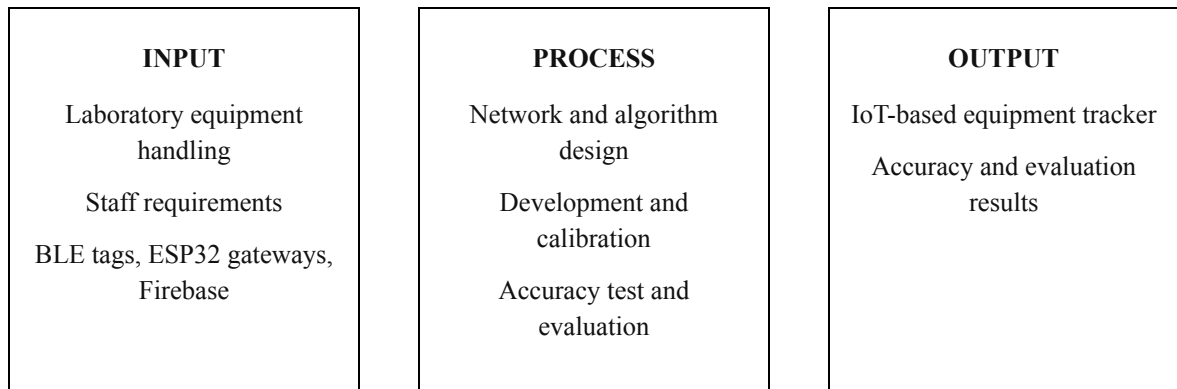
The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the equipment tracker fills.

### **Conceptual Framework**

The study follows the Input–Process–Output model shown in Figure 1.

**Figure 1**

*Conceptual Framework of the Study*



*Draft: Objectives of the Study, Scope and Delimitations and Method are being written and will be in the complete proposal manuscript.*